Black Cherry Trees: Girth vs. Volume

R fits the model · Python re-renders it · one shared dataset

The data and the fit

R's built-in `trees` dataset records girth, height, and usable timber volume for 31 black cherry trees.

```r
data(trees)
fit <- lm(Volume ~ Girth, data = trees)
slope <- coef(fit)[2]
r2 <- summary(fit)$r.squared
pred <- fitted(fit)   # plain numeric vector, for the Python bridge below
```

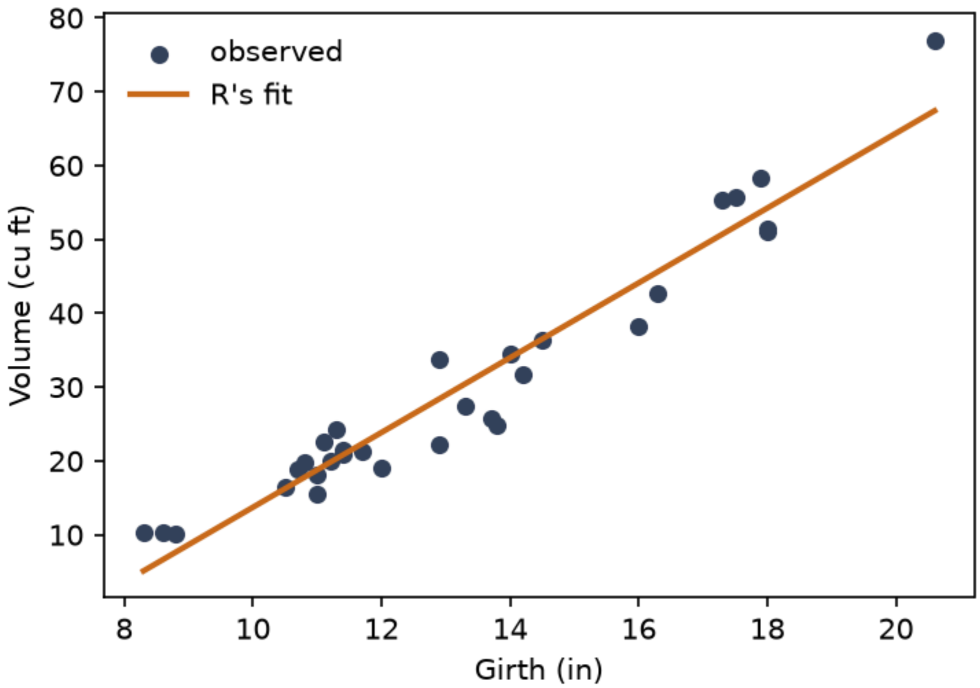
A simple linear fit puts the slope at **5.0659 cubic ft of volume per inch of girth** ($R^2 = 0.9353$) — girth alone is a strong stand-in for a tree's timber yield.

The same fit, from Python

```python
import numpy as np
import matplotlib.pyplot as plt

# r.trees and r.pred reach directly into the R chunk above -- no CSV
# round-trip, no re-fitting from scratch.
girth = np.array(r.trees["Girth"])
volume = np.array(r.trees["Volume"])
pred = np.array(r.pred)

order = np.argsort(girth)
plt.figure(figsize=(5, 3.6))
plt.scatter(girth, volume, color="#33415c", s=28, label="observed")
plt.plot(girth[order], pred[order], color="#c96a1a", lw=2, label="R's fit")
plt.xlabel("Girth (in)")
plt.ylabel("Volume (cu ft)")
plt.legend(frameon=False)
plt.tight_layout()
plt.show()
```




```
residual_spread = float(np.std(volume - pred))
```

Python isn't re-fitting anything here — `r.pred` is the fitted-values vector R just computed, read straight across the bridge and re-drawn in matplotlib's style. The residual spread computed on the Python side, 4.1125 cubic ft, is what R reports next.

Back in R

```
py_resid <- reticulate::py$residual_spread
```

Typical prediction error is about $\bar{e} = 4.1125$ cubic ft — small enough, relative to a volume range of 66.8 cubic ft, that girth alone is a serviceable field estimate for timber volume.